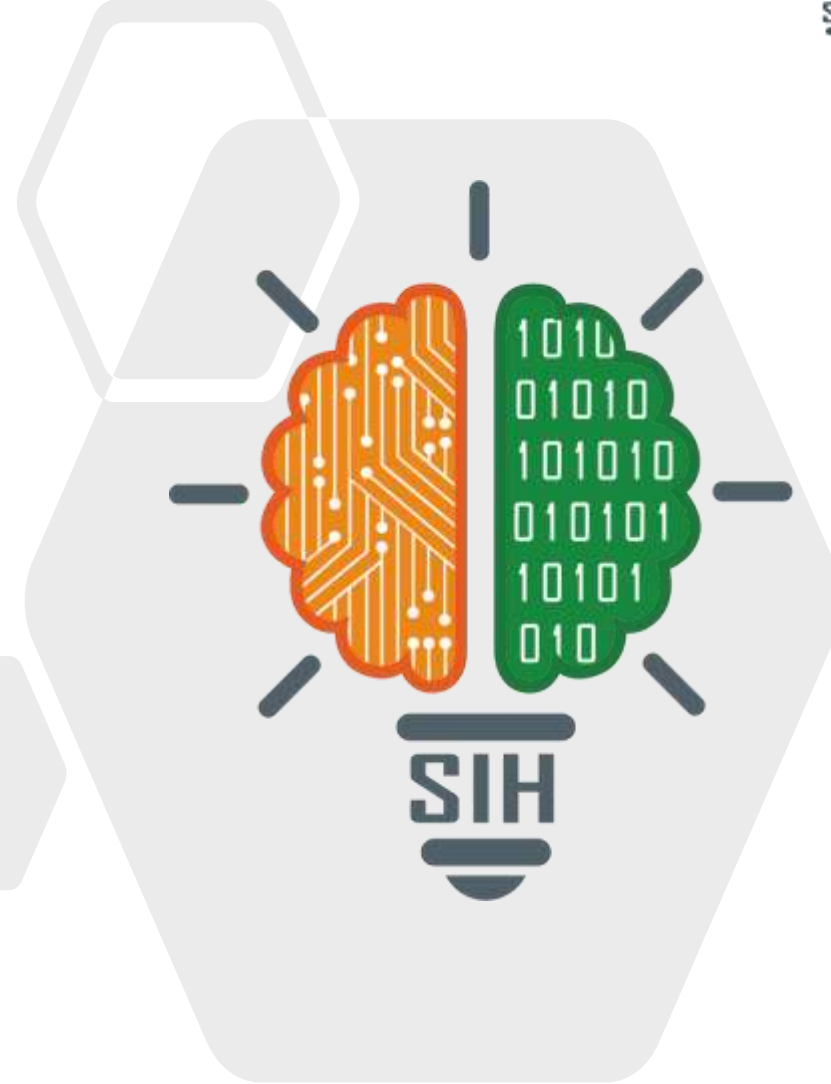


SMART INDIA HACKATHON 2025



- Problem Statement ID - 25048
- Problem Statement Title - Gamified Learning Platform for Rural Education
- Theme - Smart Education
- PS Category - Software
- Team Name – Gamify
- Team ID - 50841



Gamify – AI powered Learning Platform



PROPOSED SOLUTION

Gamify is an AI-powered learning platform that makes education engaging, inclusive, and effective, especially for students in **rural and under-resourced communities**. It combines **structured academics** with game elements to create an **interactive and enjoyable** learning environment.

- **Interactive Game-Based Lessons:** Students learn Science and Math concepts through **mini-games, simulations, and fun challenges**.
- **AI Doubt Assistant:** This feature offers instant answers to questions in **multi-language**, allowing for continuous learning without needing a teacher present all the time.
- **Gamification:** Students earn **XP, badges, and avatars** to stay motivated and consistent. **Offline-First with Cloud Sync:** This feature ensures access even in areas with limited connectivity.
- **Teacher Dashboard:** Educators can **track student progress** and provide **personalized support**, **upload study documents** as well as **videos and gamified quizzes** as well.
- **AI Proctored Career Guidance Tests:** Students can take **AI-proctored self-evaluation tests** to identify their strengths and explore which **career paths** suit them best.

How the Solution Addresses the Problem:

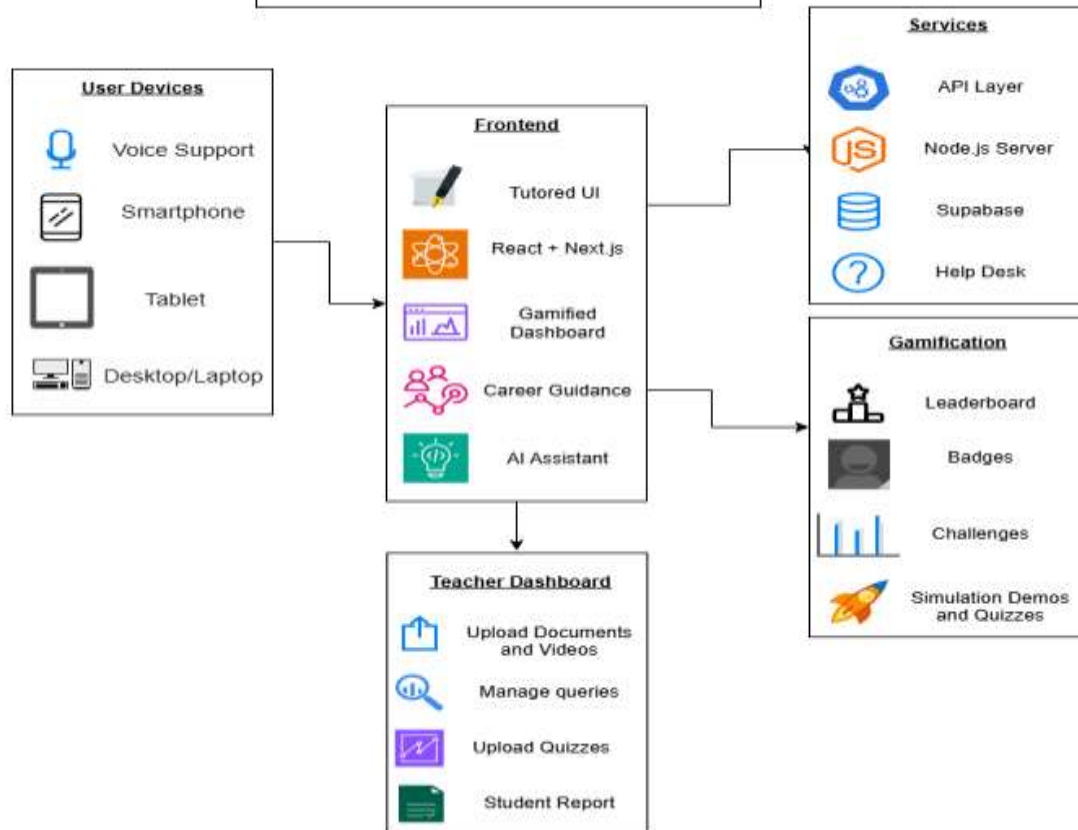
- It **bridges the gap** in quality education by making complex subjects simpler, fun, and accessible.
- It **reduces teacher dependency** with an **AI assistant** that answers questions instantly in **local languages** empowering teachers with **real-time data and analytics** to customize learning support along with **multilingual support**.
- It keeps students engaged through **rewards, streaks, and interactive challenges**, which helps **lower dropout rates**.
- It overcomes **connectivity issues** with an **offline-first design**, ensuring students can learn without interruptions in rural areas.

Innovation and Uniqueness:

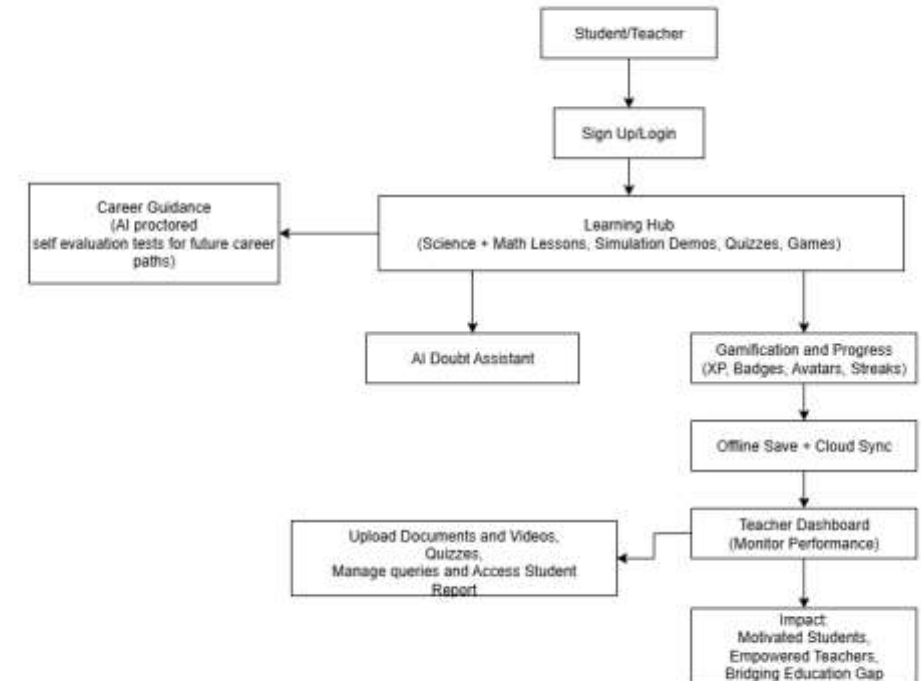
- **Localized, offline-first AI learning support** in **multi-language** and designed for rural communities.
- This offers **genuine integration of gaming and education**, not just superficial gamification.
- It is **affordable, scalable, and open-source friendly**, making it great for schools, NGOs, and government use.

TECHNICAL APPROACH

Architecture Overview



Workflow



<https://workdrive.zohopublic.in/file/swlaf29ef3b5e65c546fa8554f2ae966b7db5>



<https://github.com/SucharitaGorai/GamifyMain>



<https://gamify-sih.vercel.app/>

FEASIBILITY AND VIABILITY

Feasibility:

❖ Technical Feasibility:

- Solid Tech Stack – **React, Next. js, Node. js, Tailwind CSS** and **Supabase** for scalability and security and offline-first.
- AI Powered – **Multilingual tutoring, doubt-solving; voice & text-interaction with Google Gemini API** along with career guidance using **AI proctored self assessment tests**.
- Low-Connectivity Ready – Ideal for **rural areas** with **offline storage** and **auto cloud sync** once online.

❖ Operational Feasibility:

- User Friendly – **Easy UI, in multi regional language** for quick school student readiness.
- Empowering the Teacher – It provides option for teachers to **upload lessons/quizzes, games, videos, monitor progress and addressing queries** by accessing help desk.
- Scalable & Affordable – **Open-source, lightweight, and adaptable** for schools, NGOs, and government programs.

Challenges

Weak electricity & 4G

Language Barriers

Digital Literacy

Affordability

Engagement

Teacher Adoption

No Future Scope of Students

Strategies Infrastructure

- Offline-first with auto sync
- Lightweight app

- Multi-language
- Voice navigation & AI assistant

- Demo simulations and quizzes
- NGO/school orientation

- Open-source tools
- Shared devices

- Gamification (XP, badges)
- Local leader boards
- AI-personalized learning

- Data-driven insights
- Training sessions

- Proctored tests
- Future guidance

IMPACT AND BENEFITS



Potential Impact

- Educational Equity – It is a medium that supports access to **quality educations** irrespective of location thus ridding the educational system of discrimination.
- Improved Academic Outcomes – Students' academic performances will be better as **personalized learning, progress tracking, and gamified motivation** will be utilized.
- Dropout Reduction – The **rewards and continuous engagement** will help to keep students attending school and thereby **dropout rates will fall** dramatically.
- Digital Literacy – It **employs digital platforms** to make students familiar with technology required for the future.
- Community Development – **Local communities can build learning centers** that can support learning and interaction.
- Provides **AI-proctored self-evaluation tests**, analyzing subject-wise marks to help students identify strengths and choose the right career path.

Benefits

- For Students: Kids get involved in-it with ease and it motivates and excites them.
- For Teachers: With a teacher-friendly interface one is able to monitor students' progress.
- For Schools/NGOs: This is a solution for getting education to far-flung areas at a reasonable cost and which can be extended with ease.
- For Government: The model is in line with the state's education schemes and has a positive effect on literacy/numeracy outcomes.
- Future Career Guidance : Helps the student to find what to choose in their future to follow their future by giving AI proctored self assessment tests.

Business Model:

- B2G (Business-to-Government)
 - Partner with Odisha Government & Education Boards to integrate into school programs.
 - Licensing model where government pays per school/cluster.
- B2B (NGOs & CSR Projects)
 - NGOs and CSR initiatives adopt the platform for community centers and rural education programs.
 - Low-cost subscription or sponsorship-based model.
- Freemium Model for students:
 - Free version: Core learning content, AI assistant, and offline-first features.
 - Premium version (optional, NGO/government-sponsored): Advanced gamification, extra simulations, analytics.
- Sponsorships & Grants:
 - Collaborations with CSR funds, EdTech investors, and international education NGOs for sustainability

RESEARCH AND REFERENCES

The following are a few articles used as reference for this project:

- Saleh et al., ReWisely: Generative AI-Based Revision Platform, ICCA 2024.
- Puvvadi et al., Generative AI for Personalized Learning and Education, ICISS 2025.
- Das, K., Elementary Education in Rural Odisha, IJIRT 2025.
- The Impact of Artificial Intelligence on Personalized Learning: A Systematic Literature Review, EDUTECH Journal, 2024.
- Garg, S. & Jhajharia, P., Pedagogical Impact of Digital Education in Indian Rural Areas, 2025.
- Digital Literacy of Rural Women, GRT Journal of Education, Science & Technology, 2024.
- Elementary Education in Rural Areas of Western Orissa: Constraints and Prospects, IJCRT 2023.
- Gemini API Documentation
- NCERT Textbooks

Prototype Pictures

